Environmental pathology



Learning objective

- By the end of the lecture, you should be able to:
- Define environmental pathology, identify common environmental exposures and their pathological consequences.
- Describe the health effects of smoking and alcohol.
- Explain the negative effects of commonly abused and therapeutic drugs.
- Describe the effects and clinical consequences of radiation exposure.

Environmental Pathology

Definition: *conditions caused by exposure to chemical or physical agents in the ambient, workplace, and personal environment,* **including diseases of nutritional origin.**

undernutrition
overnutrition

Voluntary

–drug abuse
–alcohol abuse
–cigarette smoking

Involuntary

• drug, ethanol, tobacco exposure to infant and fetuses

Impact: *Environmental exposures cause ~2.3 million deaths globally each year.*

Health Effects of Climate Change

- **Greenhouse effect & rising temperatures**
  – Increased CO_2, methane, and ozone → global warming and heat waves.
- **Cardiovascular, cerebrovascular & respiratory diseases**
  – Heat stress and worsening air pollution increase morbidity.
- **Water- and food-borne infections**
  – Floods contaminate water sources → gastroenteritis, infectious disease outbreaks.
- **Vector-borne diseases**
  – Warmer temperatures expand mosquito habitats → dengue, malaria.
- **Malnutrition**
  – Crop failures due to heat → reduced food supply, especially in low-income countries.

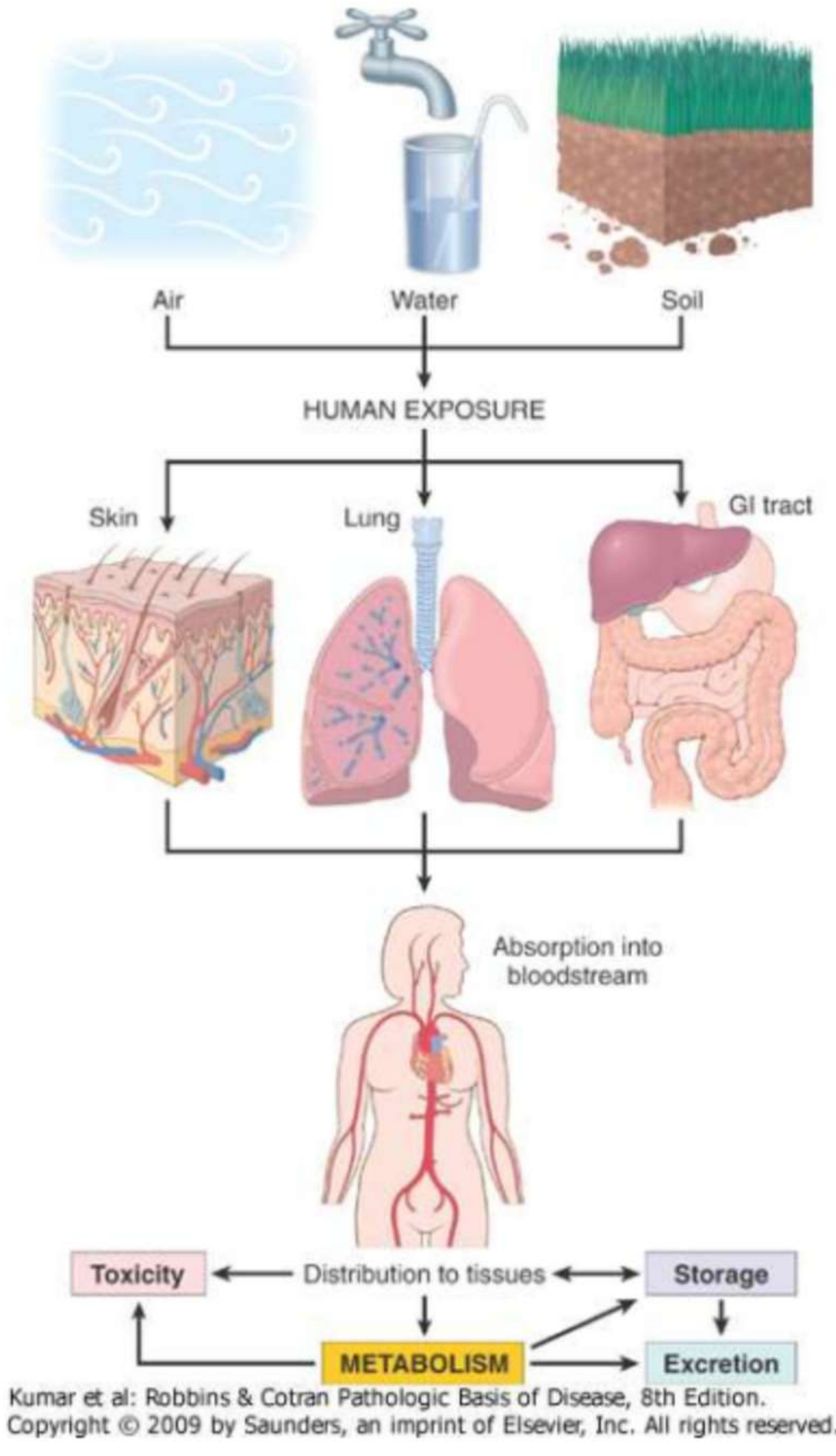
State of being diseased

Ex. -Salmonellosis -Listeriosis (monocytogenes) -Hepatitis A
-E.coli infection -Norovirus infection -Botulism (Clostridium botulinum)
-Campylobacteriosis (campylobacter)

Aedes → Dengue, yellow fever, lymphatic filarisis, Zika virus fever
Anopheles → Malaria, O'nyong'nyong virus,
Culex → Japanese encephalitis, West nile fever
Ticks → Lyme disease
Flea & lice → plague, epidemic typhus, louse-borne relapsing fever (lice)
(fleas)
-Sandflies & Blackflies → Leishmaniasis & sandfly fever (sandflies), Onchocerciasis (blackflies)

-Staphylococcal food poisoning (Staphylococcus aureus)

Human exposure to pollutants.



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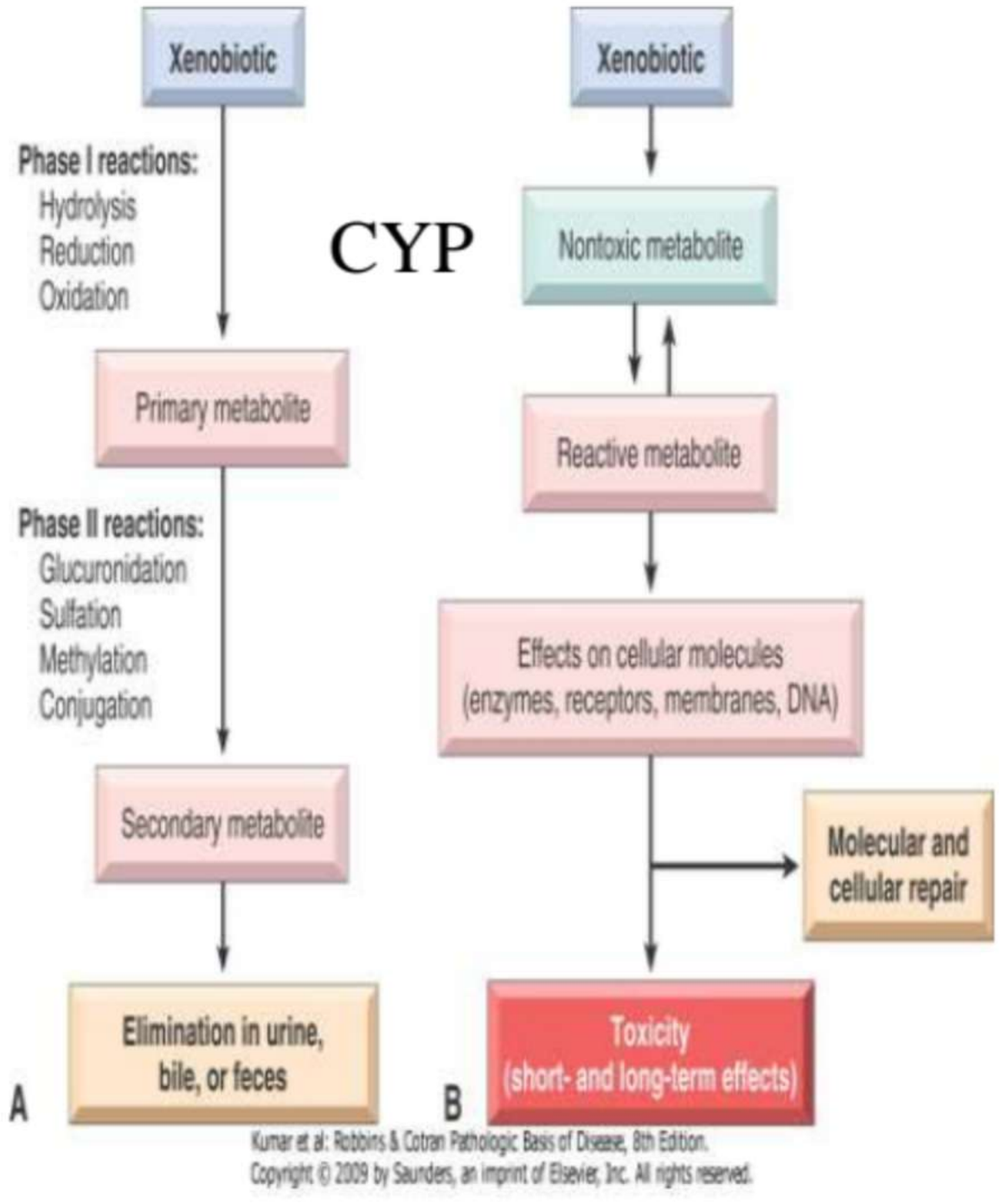
- • Pollutants in **air, water, and soil** enter the body through the **lungs, GI tract, or skin**.
  • After entering the body, they are carried in the **bloodstream** to different organs.
  • There, they may be **stored**, **excreted**, **metabolized**, or cause **toxicity**.
  • **Xenobiotics** (foreign chemicals) may be converted into **water-soluble compounds** for excretion or into **toxic metabolites**.

mechanism

Human exposure to mechanism pollutants.

- **Pathway A (Detoxification):**
  Xenobiotics (foreign chemicals) are converted into **nontoxic, water-soluble metabolites** and eliminated from the body.

- **Pathway B (Toxicity):**
  Some xenobiotics are converted into **reactive metabolites** that can damage **enzymes, membranes, or DNA**.
  If the cell cannot repair the damage, **short-term or long-term toxicity** occurs.



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Common environmental exposures

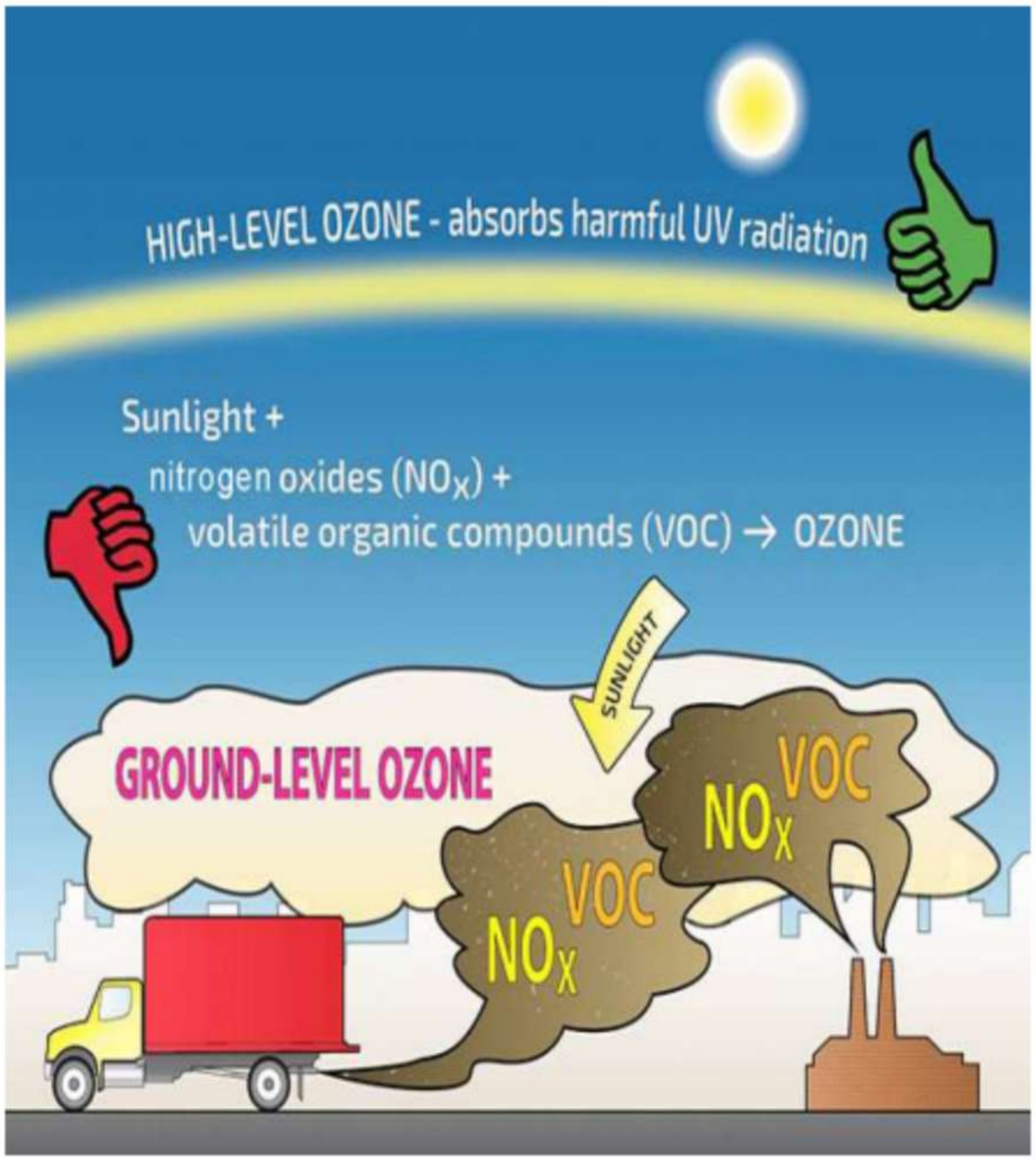
- Air pollution
- Tobacco
- Alcohol
- Drug abuse
- Therapeutic drugs
- Radiation

Air pollution

1. Ozone
2. sulfur dioxide
3. Particulates
4. carbon monoxide

Ground ozone

- Produced when sunlight reacts with motor vehicle exhaust and industrial emissions.
- Forms free radicals that injure the respiratory lining.
- Causes irritation of the upper airways, reduced lung function, and chest discomfort.
- More harmful for people with asthma, COPD, or emphysema. Chronic obstructive pulmonary disease



Carbon monoxide: Silent Killer

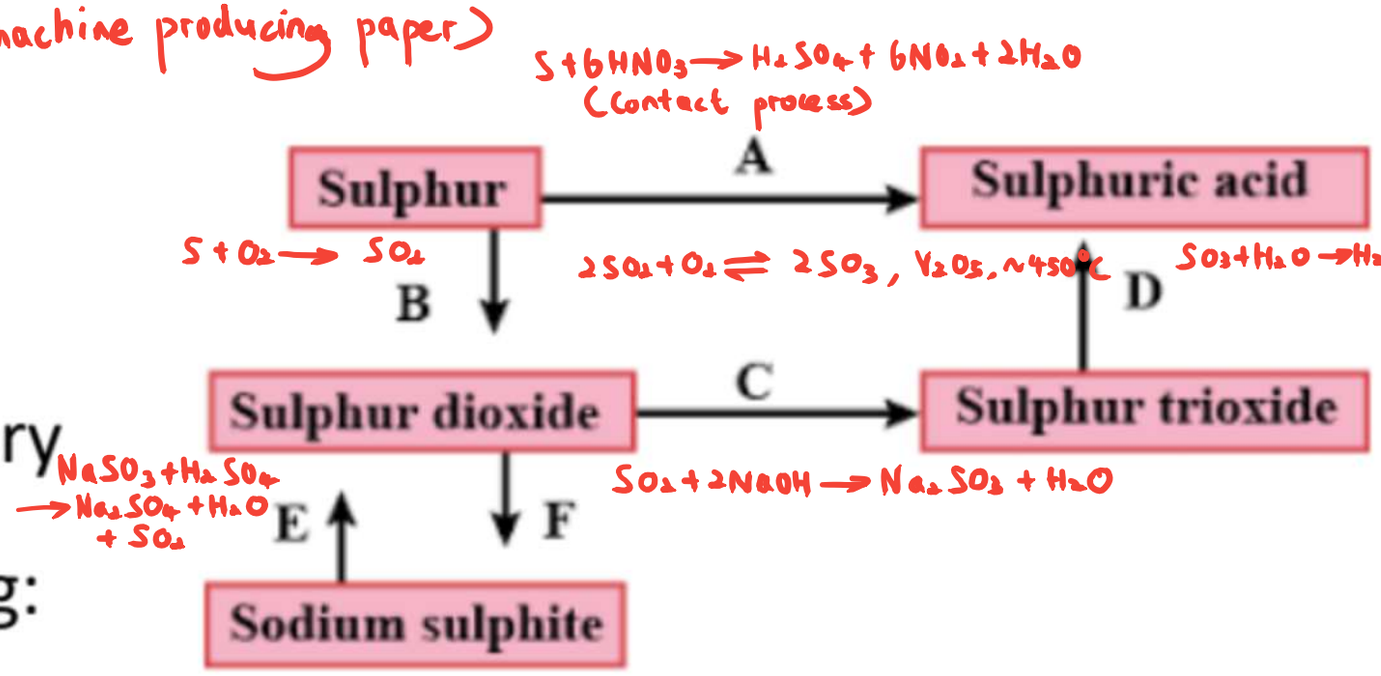
- Produced by **incomplete burning** of fuel, wood, charcoal, and cigarette smoke.
- CO binds to hemoglobin with **200 times** greater affinity than oxygen → forms **carboxyhemoglobin** that **cannot carry oxygen**.
- **20–30%** carboxyhemoglobin → **systemic hypoxia** with symptoms such as **headache, dizziness, nausea, and breathlessness**.
- **60–70%** saturation → **unconsciousness and death** are likely.



- Hyperbaric chamber → deliver Hyperbaric Oxgen Therapy (HBOT)
- involve breathing near 100% pure oxygen at air pressures two to three times higher than normal Patm.

Sulfur dioxide

- • Released from **coal- and oil-burning power plants**, **copper smelting**, and **paper mills**. (machine producing paper)
- In the air, SO_2 can be converted into **sulfuric acid** and **sulfur trioxide**.
- Highly water-soluble → absorbed in upper respiratory tract → irritate the **nose, throat, and airways**, causing:
  - – burning sensation
  - – coughing and breathing difficulty
  - – **asthma attacks** in susceptible individuals



Particulate matter (PM)

- Produced by **coal and oil power plants**, industrial burning, and **diesel exhaust**.
- PM is inhaled into the **alveoli**, where immune cells (macrophages and neutrophils) release **inflammatory mediators**.
- Causes **eye, throat, and lung irritation**, may trigger **asthma attacks**, and can worsen **heart disease**.
- PM pollution also reduces **visibility (haze)**.



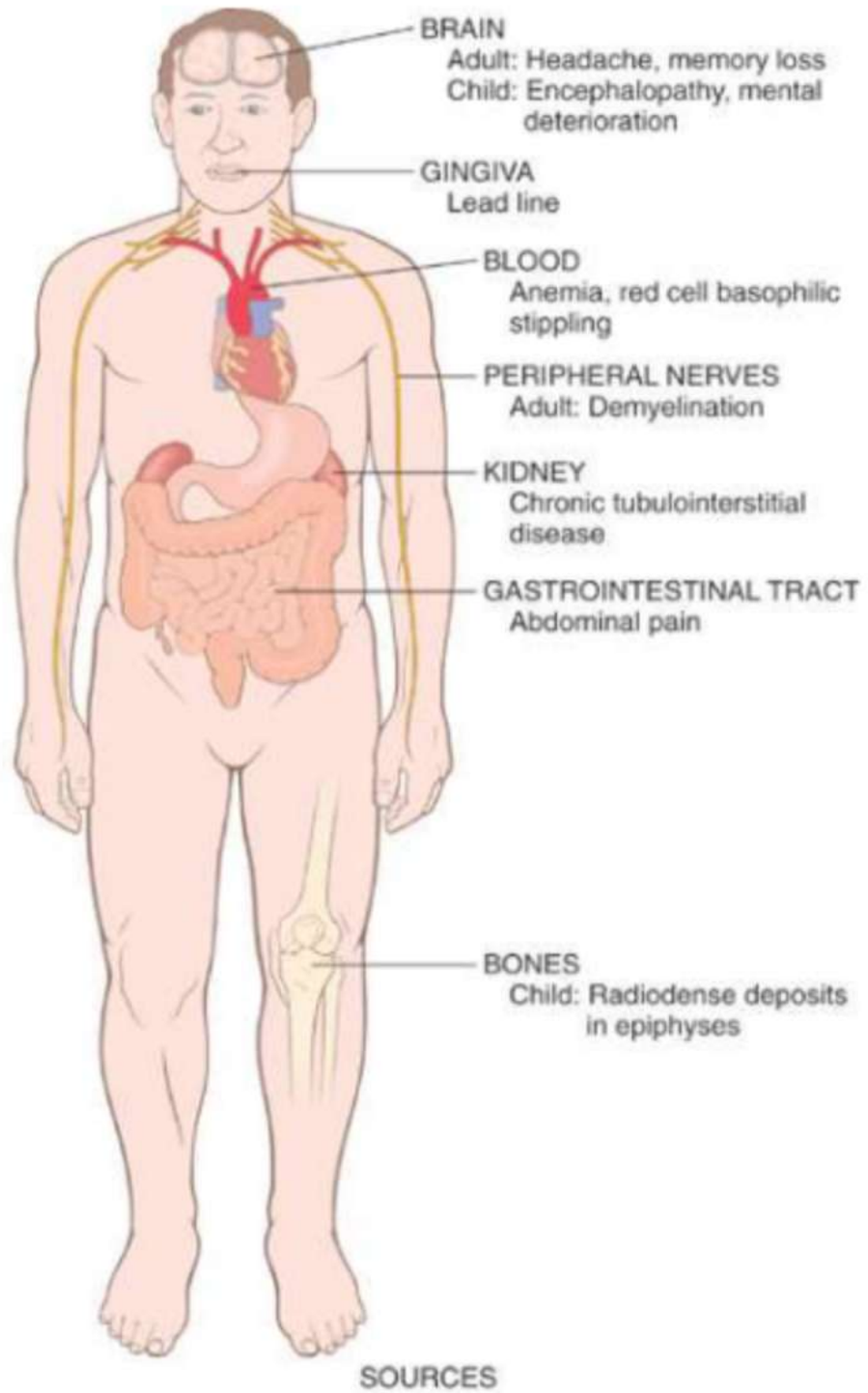
Particulate matter (PM) pollution also impacts the environment by decreasing visibility (haze).

METALS AS ENVIRONMENTAL POLLUTANTS

1. Lead
2. Mercury
3. Arsenic
4. Cadmium

Lead Exposure

- Lead accumulates in **bone and teeth**, where it competes with **calcium**.
- It interferes with **calcium-dependent neurotransmitter function**, affecting the **CNS**.
- **Effects in Adults**
  - **Peripheral neuropathy** (nerve damage).
  - **Anemia** due to impaired hemoglobin synthesis.
- **Effects in Children**
- Children absorb **>50%** of ingested lead → more lead enters the body.
- A **more permeable blood–brain barrier** makes children highly vulnerable.
- Results in **encephalopathy**, = brain dysfunction developmental delay, and brain injury.
- Lead also disrupts **cartilage remodeling** and **impairs fracture healing**. mechanism

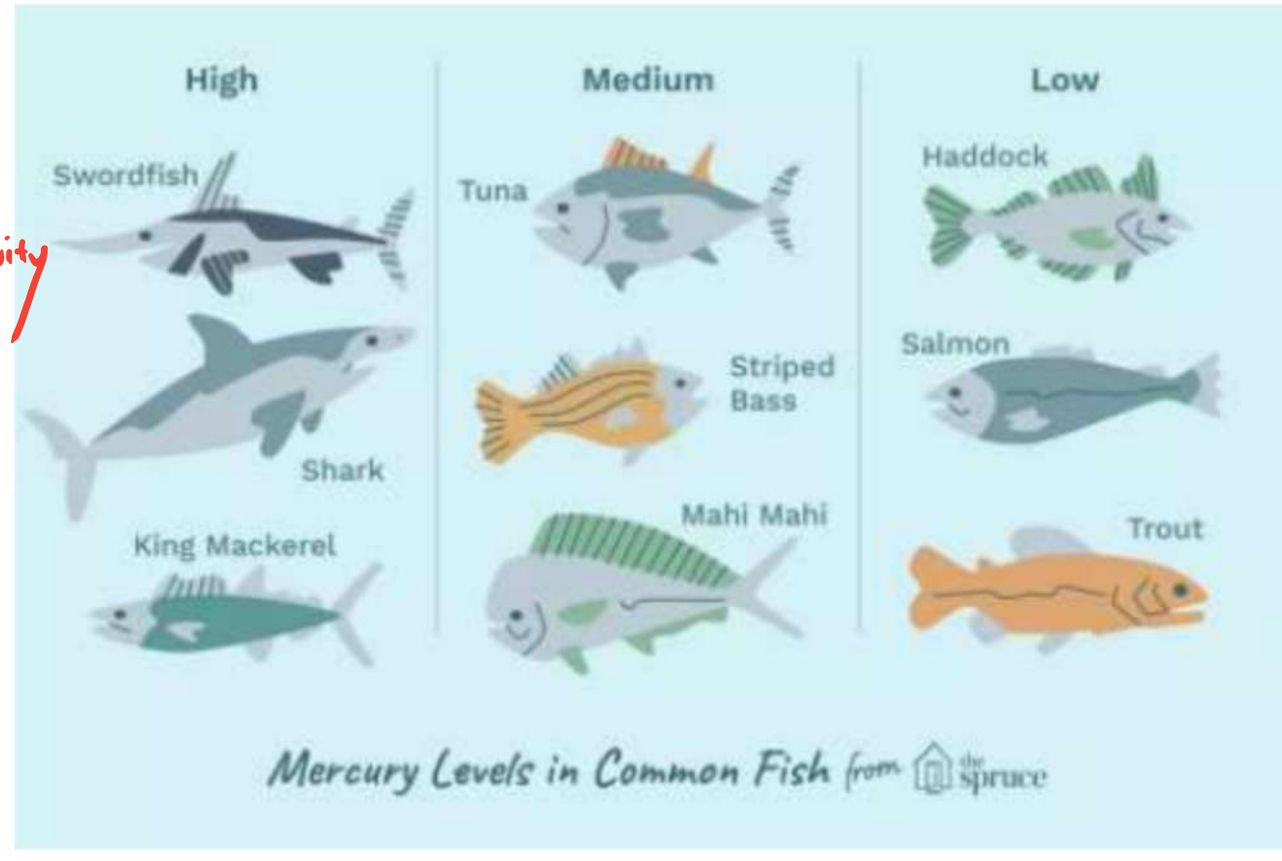


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Mercury



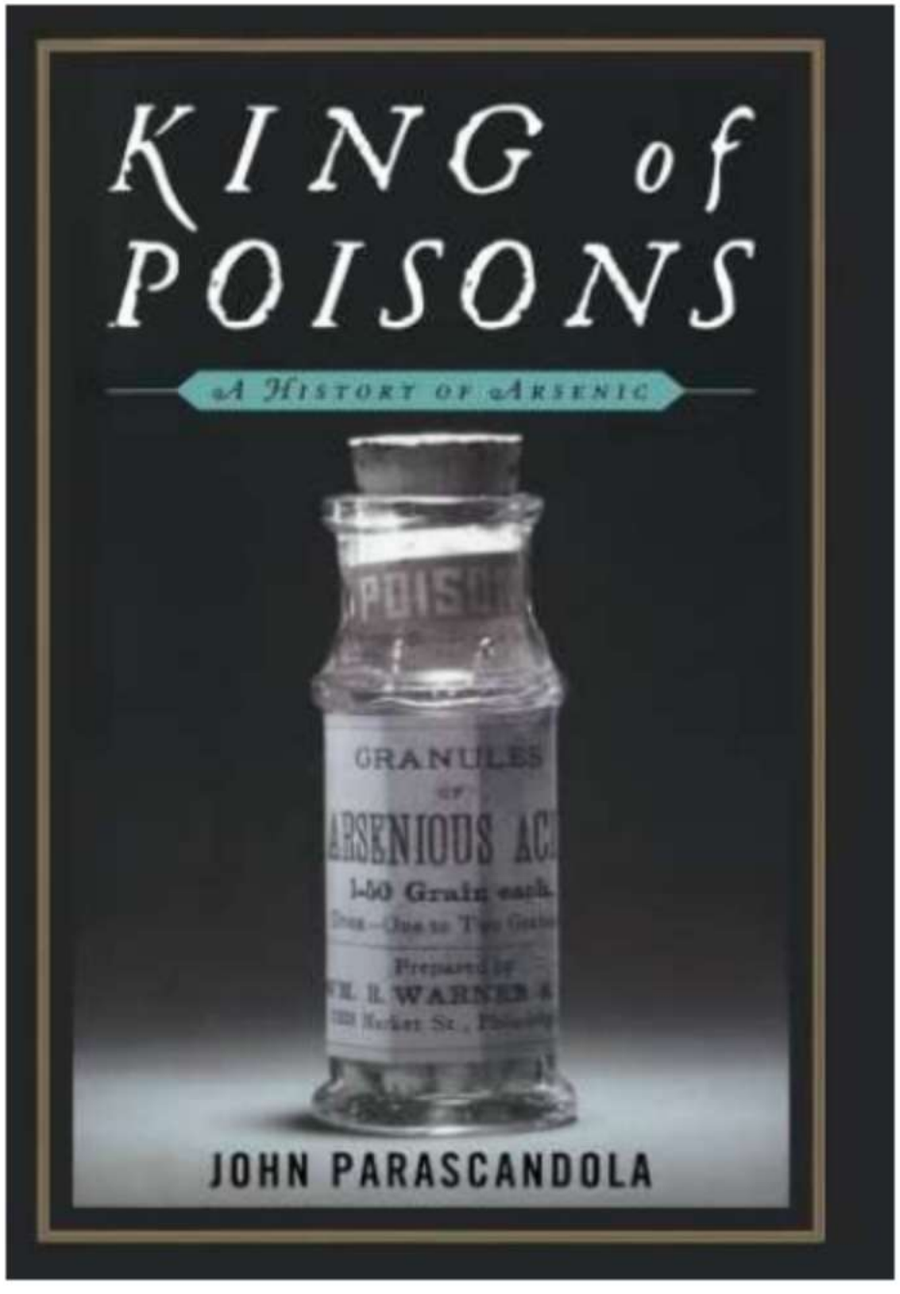
- • Mercury binds strongly to **sulfhydryl groups** in proteins → damages the **brain** and **kidneys**. mechanism
- Sources include **contaminated fish** and **mercury vapour** from dental amalgams. Silver-coloured filling material to repair tooth cavity
- The **developing fetal brain** is especially sensitive to **methylmercury**.
- Mercury accumulates in the brain, causing **neuromotor**, **cognitive**, and **behavioural** problems.
- Pregnant women should **avoid fish with high mercury levels**.



Arsenic

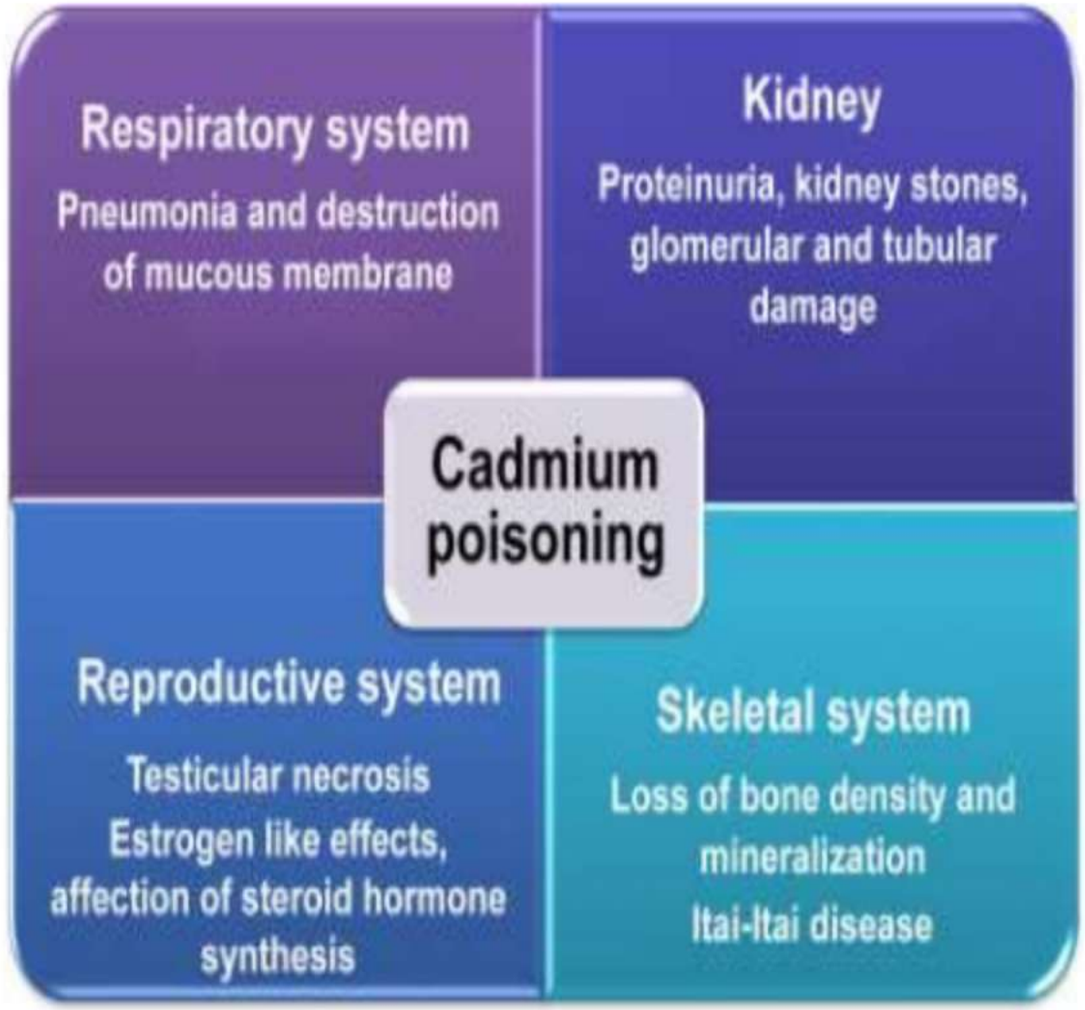
Mechanism

- Found naturally in **soil and water**; also used in **wood preservatives**, herbicides, and some traditional medicines.
- Arsenic disrupts **cellular metabolism**, mainly affecting the **GIT, heart, nervous system, and skin**.
- Causes severe **gastrointestinal**, **cardiovascular**, and **neurological** toxicity, which can be **fatal**.
- Interferes with **mitochondrial oxidative phosphorylation**, reducing ATP production.
- Chronic exposure increases the risk of **cancers** of the **lungs, bladder, and skin**



cadmium toxicity Mechanism

- Cadmium mainly damages the **kidneys** and **lungs** by increasing **reactive oxygen species (ROS)**.
- Sources include **mining**, **electroplating**, and **nickel–cadmium batteries**.
- Inhaled cadmium causes **chronic lung injury**, leading to **inflammation** and **obstructive lung disease**.
- Cadmium also causes **kidney tubular damage**, which can progress to **end-stage renal disease**.
- Long-term exposure may lead to **bone loss** and skeletal abnormalities (**Itai-itai disease**).

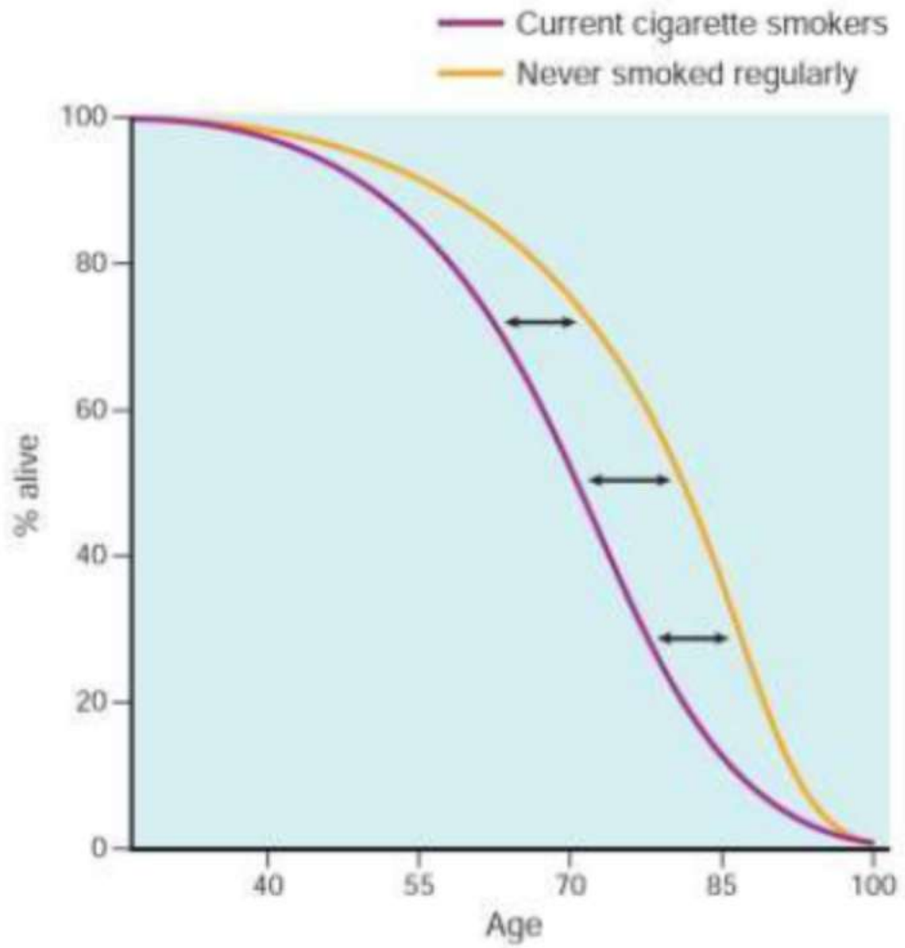


"Itai-itai" is Japanese for **"It hurts! It hurts!"**
This reflects the **severe bone pain** experienced by affected patients.

Tobacco

Mechanism

- The **most common exogenous cause of human cancer**.
- **Smoking is the most preventable cause of death.**
- Tobacco smoke contains **>7000 chemicals**, including **>60 carcinogens**.
- Causes cancers of the **lung, larynx, oral cavity, esophagus, bladder, and pancreas**.
- Smokers have a **much lower survival rate** than nonsmokers (≈50% of smokers reach age 70 vs ≈75% of nonsmokers).



The survival curve shows that smokers die earlier and at a faster rate, losing approximately 10 years of life expectancy compared to nonsmokers.

About 75% of nonsmokers are alive at age 70, only about 50% of smokers survive to 70

Examples of carcinogens in tobacco smoke

Table 9.4 Suspected Organ-Specific Carcinogens in Tobacco Smoke

Organ	Carcinogen
Lung, larynx	Polycyclic aromatic hydrocarbons 4-(Methylnitrosoamino)-1-(3-pyridyl)-1-butanone (NNK) Polonium 210
Esophagus	*N'*-Nitrosonornicotine (NNN)
Pancreas	NNK
Bladder	4-Aminobiphenyl, 2-naphthylamine
Oral cavity (smoking)	Polycyclic aromatic hydrocarbons, NNK, NNN
Oral cavity (snuff)	NNK, NNN, polonium 210

Data from Szczesny LB, Holbrook JH: Cigarette smoking. In Rom WH, editor: *Environmental and Occupational Medicine*, ed 2, Boston, 1992, Little, Brown, p 1211.

- **Tobacco smoke contains many carcinogens that act on different organs.**
  Some important examples include:
- **Polycyclic aromatic hydrocarbons (PAHs)** – cause cancers of the **lung** and **larynx**
- **Nitrosamines (NNK, NNN)** – linked to cancers of the **lung**, **pancreas**, and **oral cavity**
- **Aromatic amines** – associated with **bladder cancer**
- **Polonium-210** – a radioactive element found in tobacco, contributes to **lung cancer**
- **These chemicals explain why smoking increases cancer risk in multiple organs, not only the lungs.**

Smoking and cancer

- **Lung cancer**
  - **Male smokers are 25× more likely** to develop lung cancer than never-smokers.
- **Laryngeal cancer**
  - **Female smokers have over 100× higher risk** of laryngeal cancer — the highest risk in the table.
- **Oral cavity & pharynx**
  - Risk increases by **~6×** in smokers.
- **Bladder cancer**
  - Smokers have **4× higher risk** of bladder cancer.
- **Liver cancer**
  - Risk increases by **2–2.3×**.
- **Leukemia (AML)** (Acute myeloid leukemia)
  - Smokers have **almost 2×** the risk of acute myeloid leukemia.

Effect of smoking

- **Lungs**
- • **Lung cancer** (carcinogens: PAHs, nitrosamines)
  • **Chronic bronchitis** → irritation and mucus hypersecretion
  • **Emphysema** → inflammation & elastase-mediated tissue destruction

- **Heart**
- • Strongly linked to **atherosclerosis**, **heart attack**, and **stroke**
- **Gastrointestinal**
- • Increases risk of **oral**, **laryngeal**, and **esophageal** cancers
  • Higher risk of **peptic ulcers**

Effect of smoking

- Effects on Fetus
  - Causes fetal hypoxia
  - low birth weight,
  - prematurity,
  - increased spontaneous abortion
- Infants of mothers who stop smoking before pregnancy have normal birth weights.



Back

Second hand smoking



- Increases risk of:
  - **Myocardial infarction**
  - **Lung cancer**
  - **Ischemic heart disease**
  - **Sudden infant death syndrome** (maternal smoking)
- **Children living with smokers have more respiratory illnesses, including asthma.**

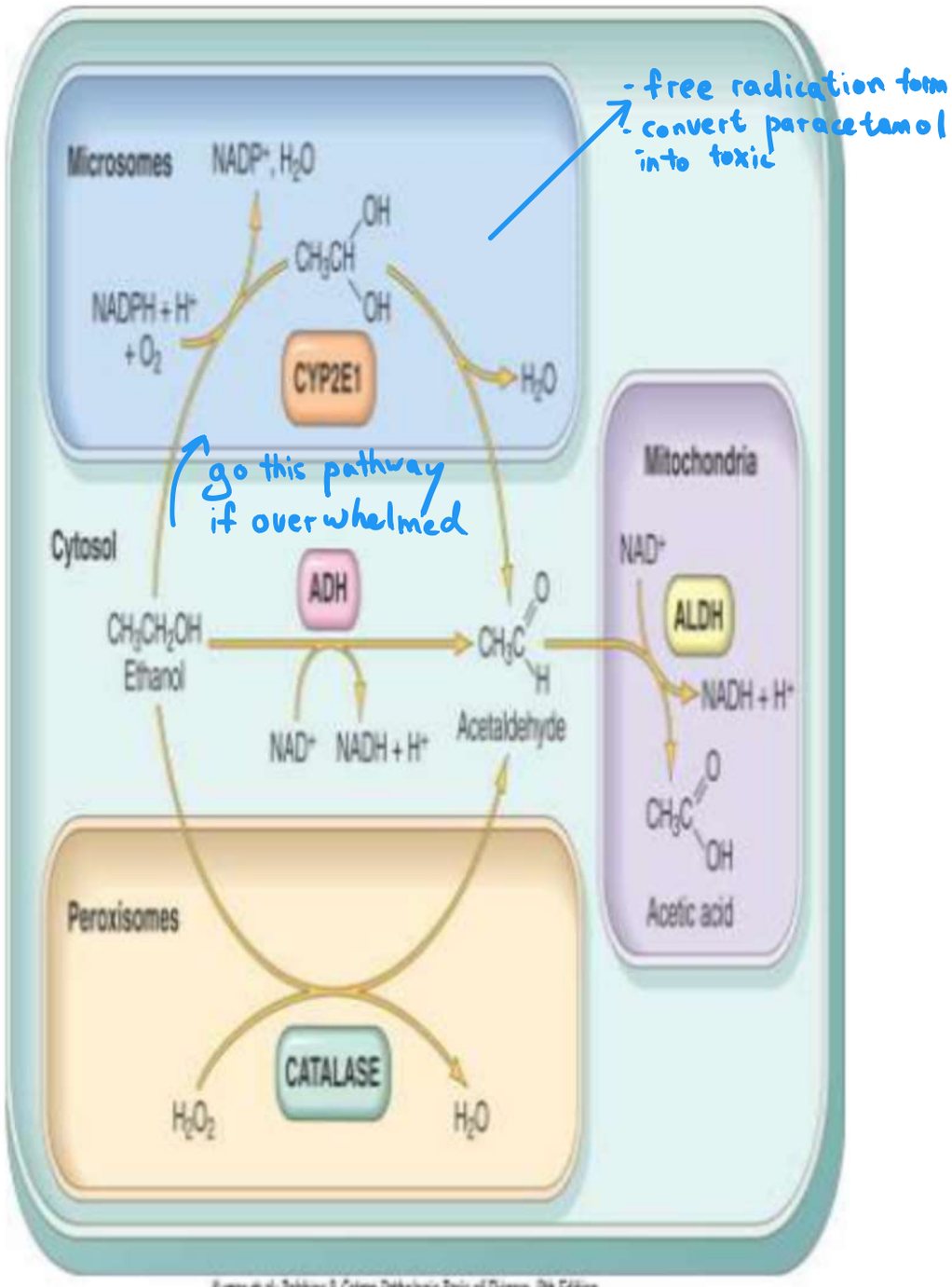
Electronic nicotine delivery systems (ENDS) Mechanism

- • ENDS use has **rapidly increased** over the past decade.
  • In the USA, ENDS are now **more popular than cigarettes** among high school students.
  • The **long-term health effects** of ENDS are still uncertain.
  • Many smokers become **dual users** of both cigarettes and ENDS rather than quitting.
  • Young people addicted to **nicotine** through ENDS may later switch to cigarettes.



Alcohol

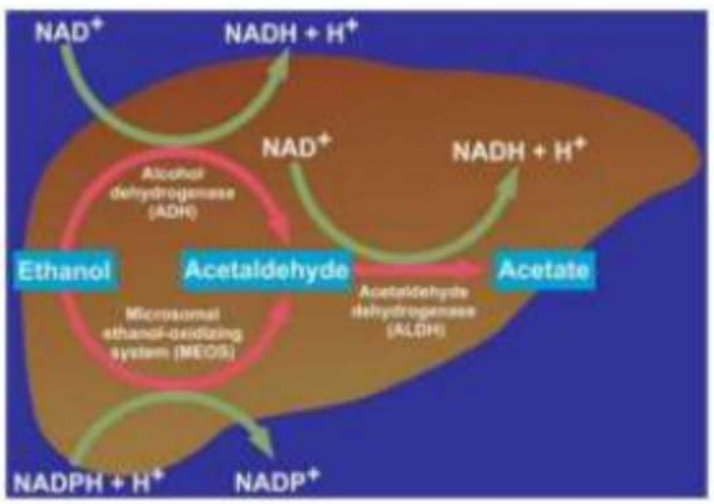
- Alcohol is the **most widely abused** substance.
- Its toxic effects come mainly from **acetaldehyde**, the first metabolite of ethanol.
- Most ethanol is metabolized in the liver by **alcohol dehydrogenase (ADH)**.
- At high alcohol levels, an additional system (**MEOS/CYP2E1**) helps metabolize ethanol.
- Induction of CYP2E1 explains why chronic alcohol users are more sensitive to **other drugs and toxins**.
- Example: Chronic alcohol intake induces CYP2E1 → increases conversion of paracetamol into toxic NAPQI → higher risk of acute liver failure even at moderate doses.



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Acetaldehyde

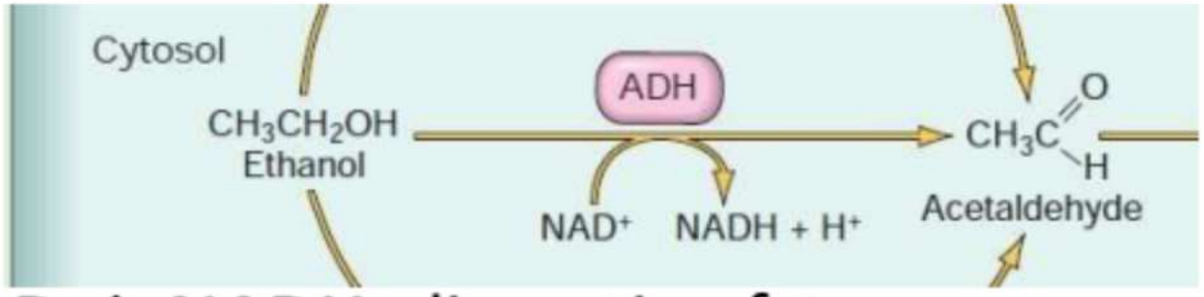
- **Acetaldehyde**
- Acetaldehyde is the **toxic intermediate** formed during alcohol metabolism.
- It is responsible for many acute effects of alcohol (flushing, headache, nausea).
- Alcohol metabolism varies between individuals due to differences in the enzymes **ADH** and **ALDH**. (Aldehyde dehydrogenase) (Alcohol dehydrogenase)



- **ALDH2 Deficiency**
- About **50% of East Asians** have reduced activity of **ALDH2**.
- This causes **acetaldehyde accumulation** after drinking.
- Symptoms: **flushing**, **nausea**, **tachycardia**, **headache**, **hyperventilation**.
- Individuals with the ALDH2*2 variant **cannot tolerate alcohol** well.

Mechanism

Effects of alcohol on Liver



脂肪肝

- **1. Fatty Liver (Steatosis)**
- Alcohol metabolism converts **NAD → NADH**, disrupting fat metabolism.
- Low NAD means **fat cannot be oxidized**, so it **accumulates in the liver**.
- Result → **fatty change** in chronic drinkers.
- **2. Alcoholic Hepatitis** Mechanism
- Alcohol induces **CYP2E1**, producing **reactive oxygen species (ROS)**.
- ROS → **membrane damage**, inflammation, hepatocyte injury.
- Acute hepatitis → Alcoholic cirrhosis → hepatocellular carcinoma

Effects of alcohol on Nervous System

Alcohol replaces food intake → ↓ Food intake → Vitamin deficiencies → Thiamine deficiency: (= vitamin B1)

WERNICKE SYNDROME

degeneration of nerve cells, atrophy of the cerebellum (ataxia) ophthalmoplegia and nystagmus

ophthalmoplegia: paralysis of 1/more muscles control eye movement

nystagmus: uncontrolled, involuntary eye movements

Korsakoff syndrome

severe memory loss

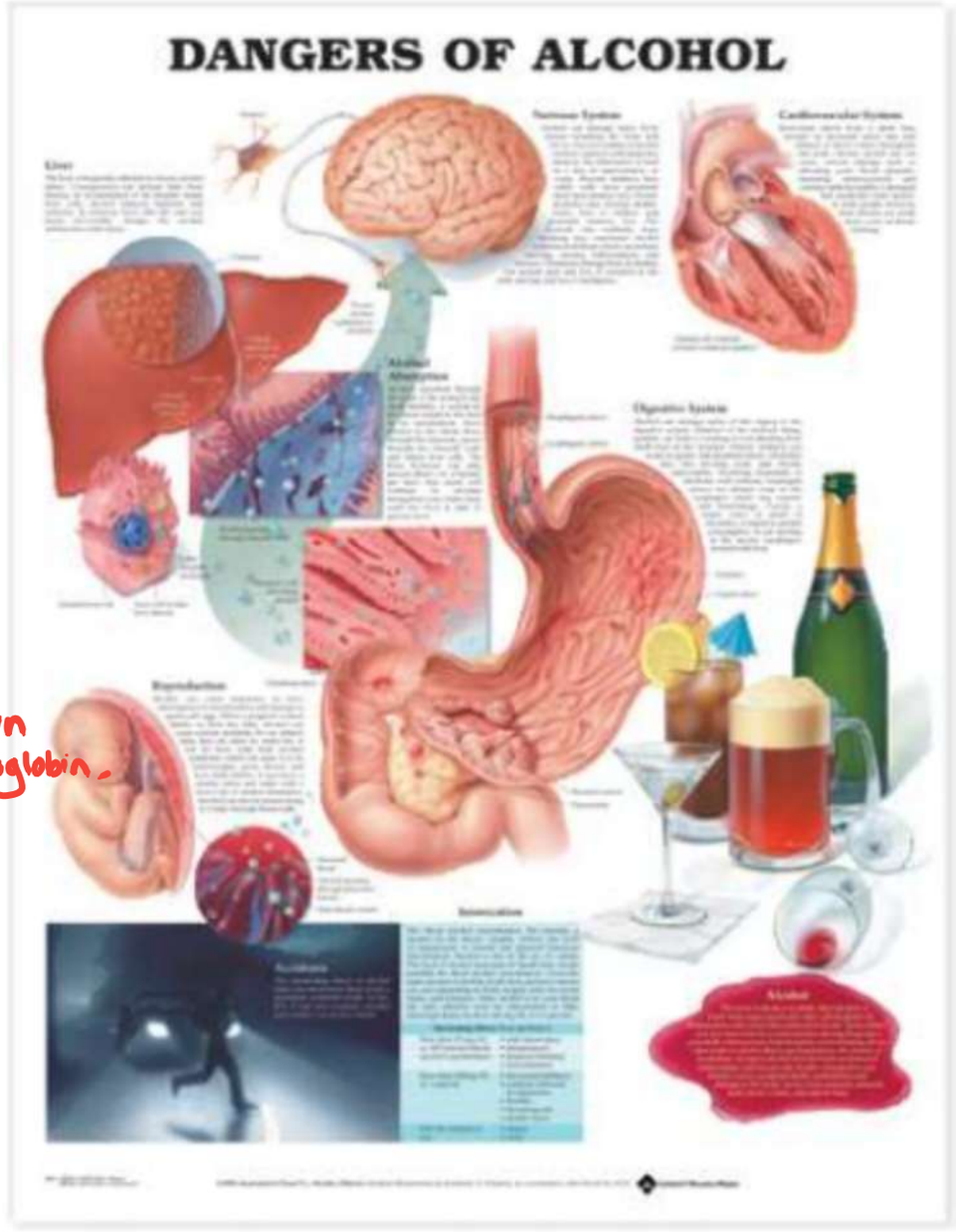
Alcohol = CNS depressant → affects subcortex → cortex → medulla → possible respiratory arrest.

Alcohol Effects on Other system

- **CVS:**
  - Dilated cardiomyopathy,
  - Hypertension
- **GIT:**
  - acute gastritis,
  - pancreatitis
- **Skeletal muscle:**
  - Rhabdomyolysis

= damaged skeletal muscle breaks down rapidly and spill its contents (eg. myoglobin, electrolytes (K^+)) into bloodstream

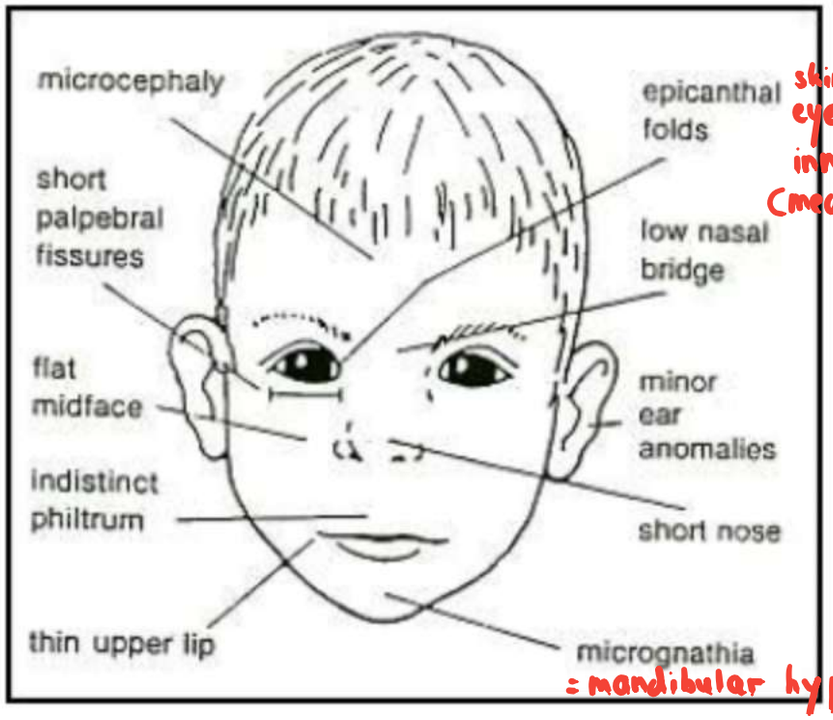
- **Reproductive system:**
  - testicular atrophy
  - spontaneous abortion



Fetal alcohol syndrome

- consumption during the first trimester of pregnancy is particularly harmful
- characterized by
  - growth and developmental defect;
  - facial dysmorphology,
  - malformation of CNS, CVS and genitourinary system

= urogenital
= related to urinary & reproductive system

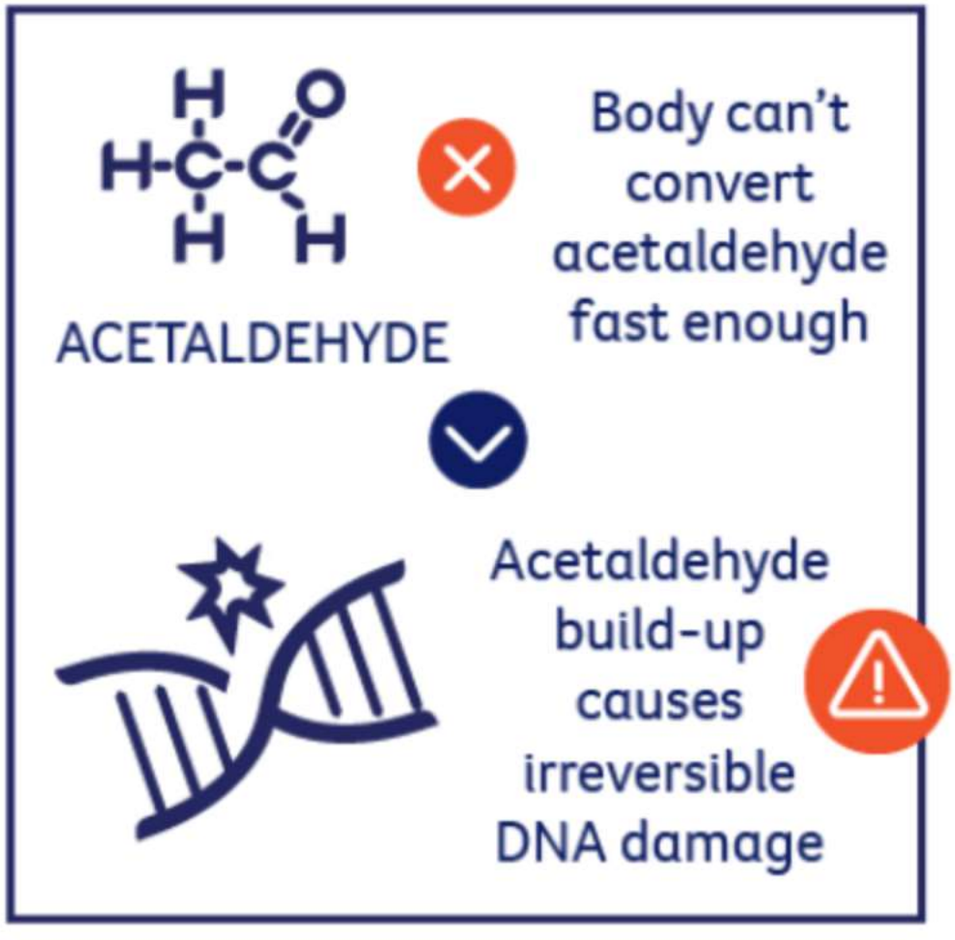


skin fold of upper eyelid that covers inner corner (medial canthus) of eye

= mandibular hypoplasia

Alcohol and Cancer

- **Alcohol is causally linked to cancers of:**
  - Oral cavity, pharynx, larynx
  - Oesophagus (SCC) squamous cell carcinoma
  - Colon & rectum
  - Liver & intrahepatic bile ducts
  - Female breast
- **Carcinogenic agent:** Acetaldehyde
- **Acetaldehyde** is toxic and can lead to irreversible DNA damage, which can lead to cancer.



Common Drug Of Abuse

cocaine,

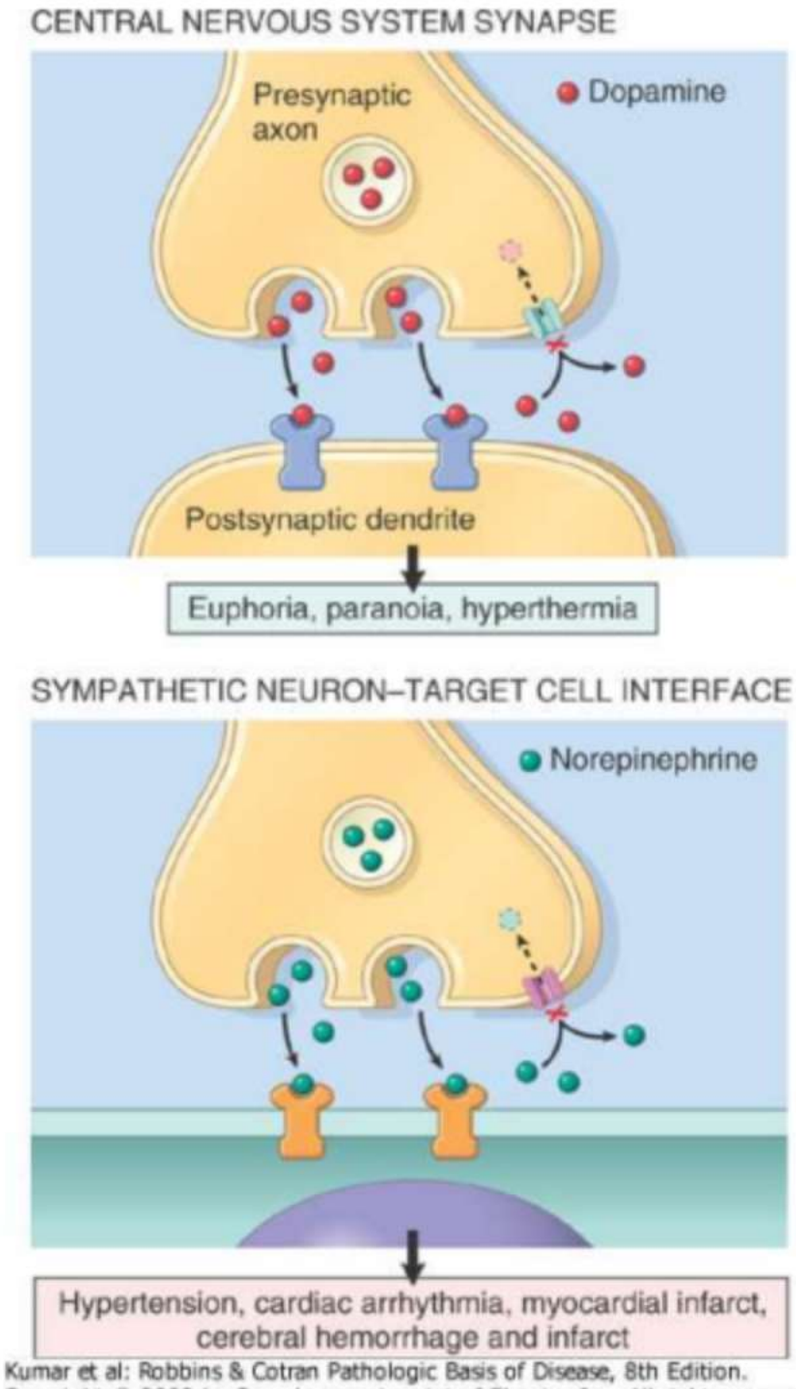
heroin,

Amphetamines

Cocaine: Euphoria to arrhythmia

heartbeat rate/rhythm got problem

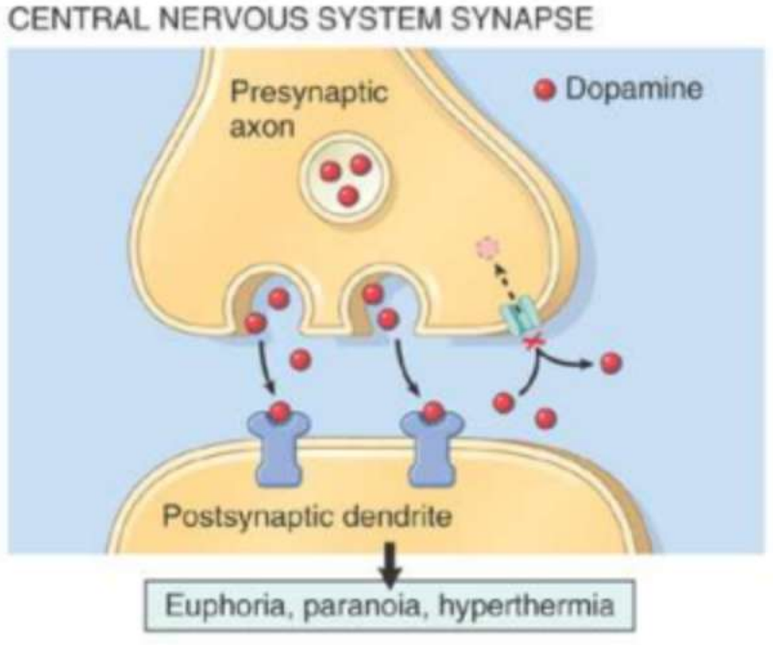
- Cocaine is a powerful stimulant that produces intense euphoria by blocking the reuptake of dopamine and norepinephrine.
- Excess neurotransmitters → hyperstimulation → euphoria, hyperactivity, paranoia 被害妄想
- Sympathetic effects:
  - Tachycardia
  - Hypertension
  - Peripheral vasoconstriction
- Overdose → seizures, arrhythmia, respiratory arrest

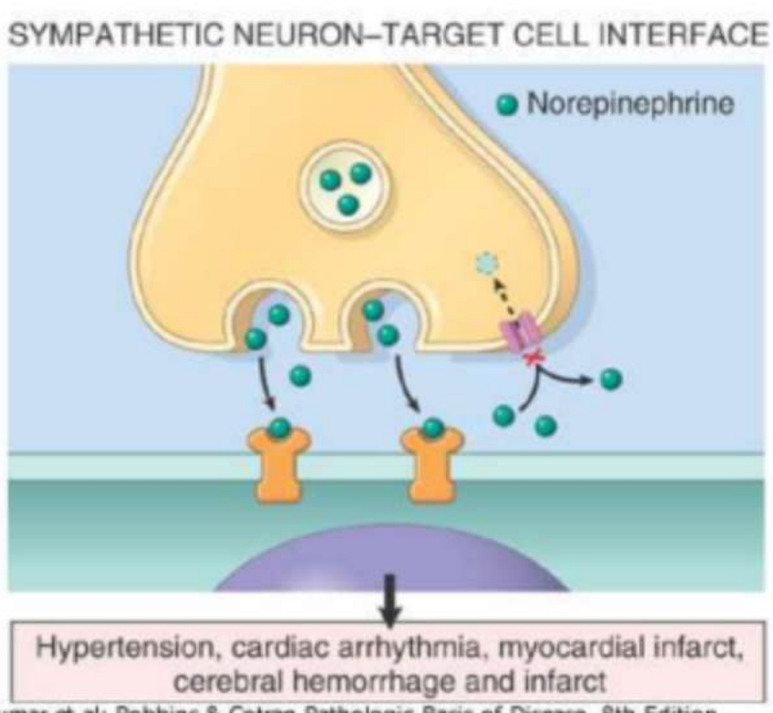


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Cocaine: Cardiovascular Toxicity

- Intense coronary vasoconstriction
- ↑ Platelet aggregation → thrombus formation
- ↑ O_2 demand, ↓ coronary flow → myocardial ischemia / infarction
- Chronic use accelerates atherosclerosis
- Can cause cerebral hemorrhage and infarct



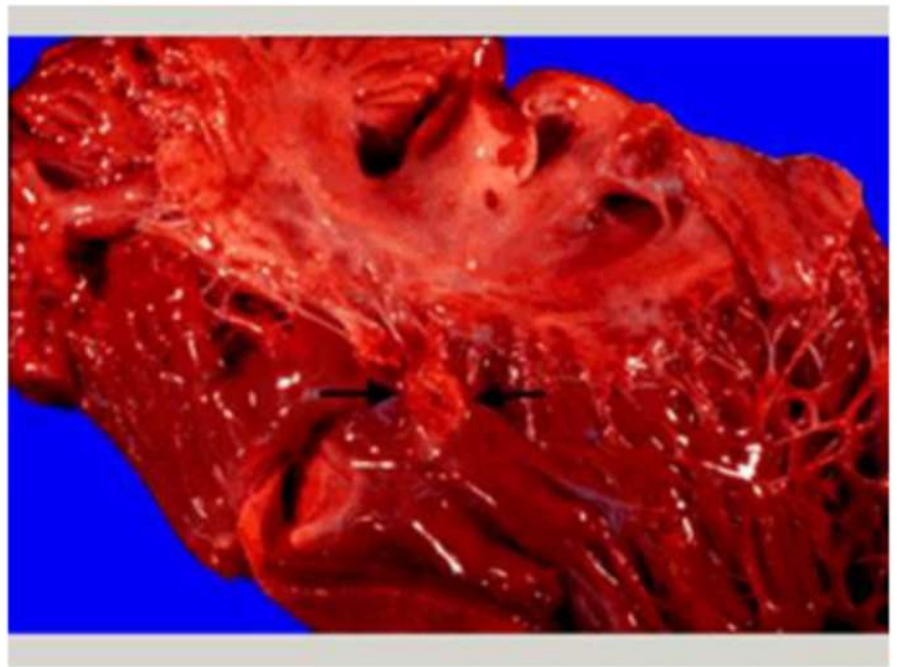


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Heroin: Infections, Endocarditis & Sudden Death

- usually self-administered intravenously or subcutaneously
- Infectious complications are common
- common sites are the skin and subcutaneous tissue, heart valves, liver, and lungs.
- right-sided heart valves endocarditis, esp the tricuspid caused by *S. Aureus* Mechanism
- Viral hepatitis and HIV is acquired by the sharing of dirty needles.
- Sudden death: overdose, especially when heroin is contaminated with potent opioids (fentanyl, carfentanil)
  - 50~100 times potent than morphine
  - 50 times than heroin
  - 10000 than morphine
  - 100 than fentanyl





Amphetamines: Dance to death

street name for MDMA (3,4-methylenedioxymethamphetamine)

- *ecstasy*.
- Methamphetamine increases dopamine release and reduces glutamate signaling → CNS stimulation, users may be unable to stay still ("constant movement").
- Methamphetamine produces a feeling of euphoria, which is followed by a "crash." (withdrawal syndrome)
- Long-term use → violent behavior, confusion, paranoia, and hallucinations
- **Adverse effect:** Excess sweating due to intense sympathetic stimulation; muscle contraction → **hyperthermia → organ failure → death**

body temp rise above normal because body absorb/generate more heat than released

- Hyperthermia may cause rhabdomyolysis → acute kidney injury → death.
- Amphetamines can cause severe hypertension → intracerebral hemorrhage or stroke.
- Chronic use leads to cardiomyopathy and arrhythmias.

Types:
- dilated ~ - restrictive ~
- hypertrophic ~ - arrythmogenic right ventricular cardiomyopathy (ARVC)

(muscle tissue in right ventricle replaced by fat and scar tissue) → irregular heart rhythms

Therapeutic Drugs

Environmental Pathology

Oral contraceptives

- Combined hormonal contraceptives are classified as carcinogenic to humans (Group 1), while progestogen-only contraceptives are considered possibly carcinogenic (Group 2B). according to International Agency for Research on cancer (IARC)
- Current or recent users of combined OCPs have a higher risk of breast, cervical, and (in low-HBV regions) liver cancer.
- Associated adverse effects:
  - Venous thrombosis and pulmonary embolism
    - OCP increases risk of thrombosis
    - increases in C-reactive protein and coagulation factors and reduction in anticoagulants (protein S and anti-thrombin III).
  - CVS disease
    - Myocardial infarction risk is higher IF > 35 years + smoking
  - Hepatic adenoma: benign, usually With prolonged use
  - Ovarian cancer: OC protect against Ovarian Cancer

Menopausal Hormone Therapy (previously known as Hormone replacement therapy)

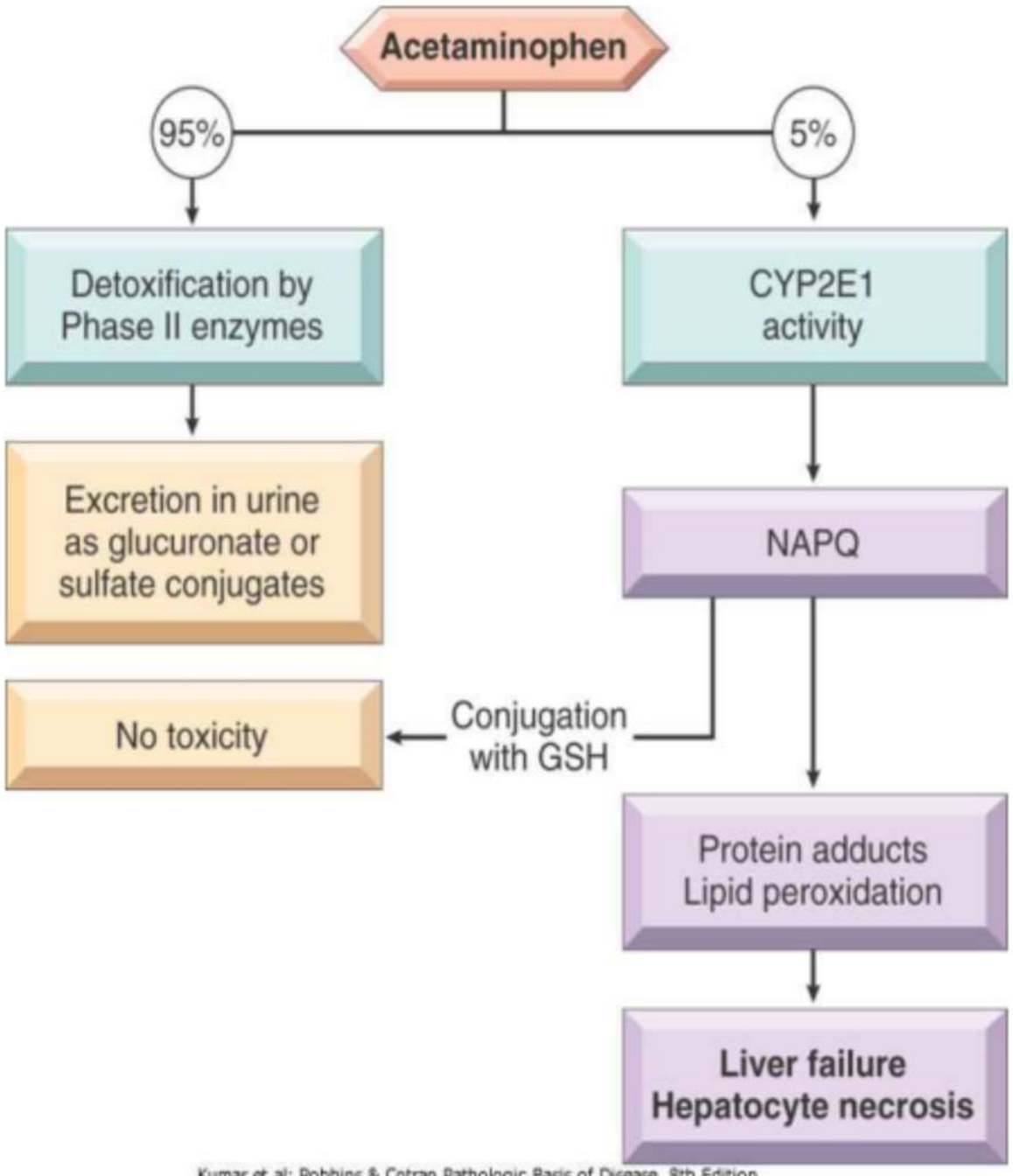
- Endometrial cancer and hyperplasia: Risk is high with unopposed Oestrogen therapy
- Breast cancer: increased risk with prolonged use
- Venous thrombosis and pulmonary embolism: elevated twofold in MHT users

100% in risk

- MHT has a protective effect on the development of atherosclerosis and coronary disease in women under age 60
- Risks and benefits associated with MHT and OCP must be evaluated individually

Acetaminophen / Paracetamol

- It causes 50% of acute liver failure in US; 30% mortality
- In Malaysia, Acetaminophen poisoning : 29.0% of all drug poisoning; 60.0% suicide attempts, and 33.3% accidental ingestions.
- At therapeutic doses: 95% of acetaminophen undergoes detoxification in the liver by phase II enzymes and is excreted in the urine.
- 5% metabolized through the activity of CYPs to NAPQI, a highly reactive metabolite.
- NAPQI is normally conjugated with glutathione (GSH), but when taken in larger doses unconjugated *NAPQI accumulates and causes hepatocellular injury leading to centrilobular necrosis and liver failure*.
- Tx: *N-acetylcysteine, which restores GSH*
- *Remember what happens when someone drinks alcohol and ingest paracetamol?*



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Aspirin (acetylsalicylic acid)

- Aspirin overdose may result from accidental ingestion of a large number of tablets by young children; in adults, overdose is frequently suicidal.
- Can cause:
  - Respiratory alkalosis followed by metabolic acidosis
  - Chronic aspirin toxicity (salicylism): headache, dizziness, tinnitus, convulsions, coma
  - Acute erosive gastritis → GI bleeding / ulcer
  - Bleeding tendency: aspirin inhibit platelet aggregation (aspirin acetylates platelet cyclooxygenase and irreversibly blocks the production of thromboxane A2, an activator of platelet aggregation
- Acute salicylate overdose causes alkalosis as a consequence of the stimulation of the respiratory center in the medulla.
- This is followed by metabolic acidosis and accumulation of pyruvate and lactate, caused by uncoupling of oxidative phosphorylation and inhibition of the Krebs cycle.
- Metabolic acidosis enhances the formation of non-ionized forms of salicylates, which diffuse into the brain and produce effects from nausea to coma.

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Aspirin + PCM

- Aspirin + PCM → chronic intake = renal papillary necrosis OR **analgesic nephropathy**
- PGE_2 maintains renal blood flow; its synthesis depends on cyclooxygenase.
- Aspirin inhibits COX → ↓ PGE_2 → renal vasoconstriction → ↓ blood flow.
- In renal cells, paracetamol is converted to into *N*-acetyl-*p*-benzoquinoneimine (NAPQI).
- NAPQI depletes glutathione via non-enzymatic conjugation to glutathione, a naturally occurring antioxidant.
- With depletion of glutathione, cells of the kidney become particularly sensitive to oxidative damage.

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Radiation injury

Radiation

Energy that travels as waves or high-speed particles.

Ionizing

X-ray,
gamma ray,
cosmic waves

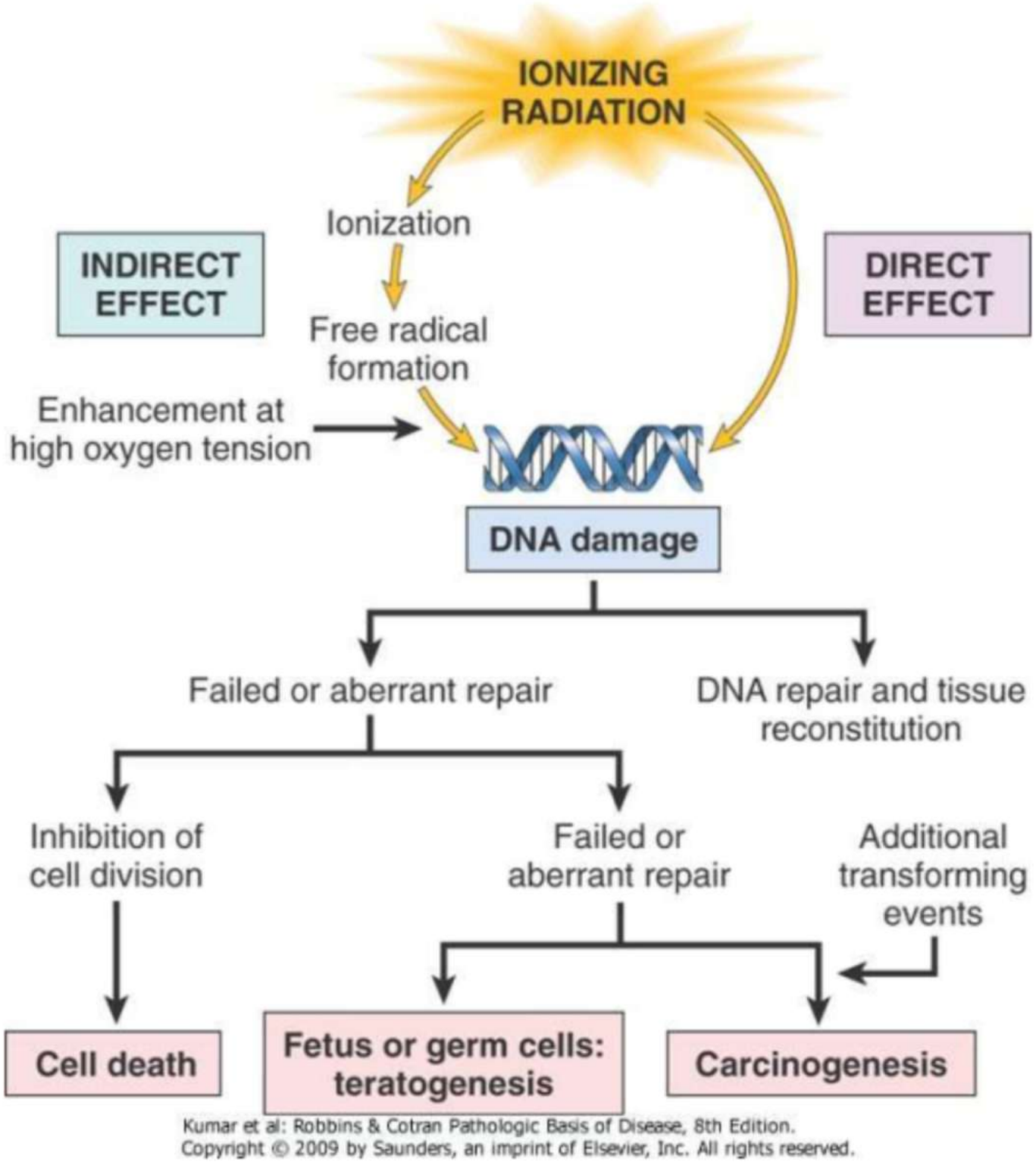
- Removes tightly bound electrons → ion formation
- Causes electron cascade → DNA damage

Non-ionizing

Microwave,
infrared,
UV light (mostly non ionizing

- Causes vibration/heat
- Not enough energy to remove electrons

- DNA damage → mutations → **cancer**
- Irreparable DNA injury → **cell death**



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Radiation injures DNA through direct ionization or indirect free radical formation. Correct repair allows recovery; failed repair causes mutations → cancer, teratogenesis, or cell death.

Factors involved in ionizing radiation effects

- Dose rate:
  - Single high dose causes more injury than the same dose divided over time
- Cell cycle:
  - Cells in G2 and M phase are most radiosensitive
- Cell types:
  - rapidly dividing cells e.g. hematopoitic cells, germ cells, GI epithelium
- *Oxygen effects and hypoxia*.
  - **Free radical formation (from water radiolysis) is the main mechanism of DNA damage.**
  - Oxygen converts radiation-induced free radical DNA damage into permanent, non-repairable lesions. Therefore, **Hypoxic tissues are more resistant to radiation therapy.**
- *Vascular damage*.
  - Endothelial injury → vessel narrowing/occlusion → impaired healing and ischemic atrophy

Cellular mechanism of radiation injury

- Acute effect:
- Fibrosis
- Carcinogenesis

Acute effect

Acute Effects on Hematopoietic and Lymphoid Systems.

- **Lymphoid tissue:**
  – Lymphocytes are *extremely radiosensitive* → immediate lymphopenia
  – Shrinkage of nodes, spleen, thymus
- **Bone marrow:**
  – Hematopoietic precursors very sensitive to radiation
  – High dose → **marrow aplasia**
  – Very high dose → **permanent aplastic anemia**
  – Lower dose → transient aplasia

These tissues contain labile cells → rapid turnover → highest radiosensitivity.

Fibrosis

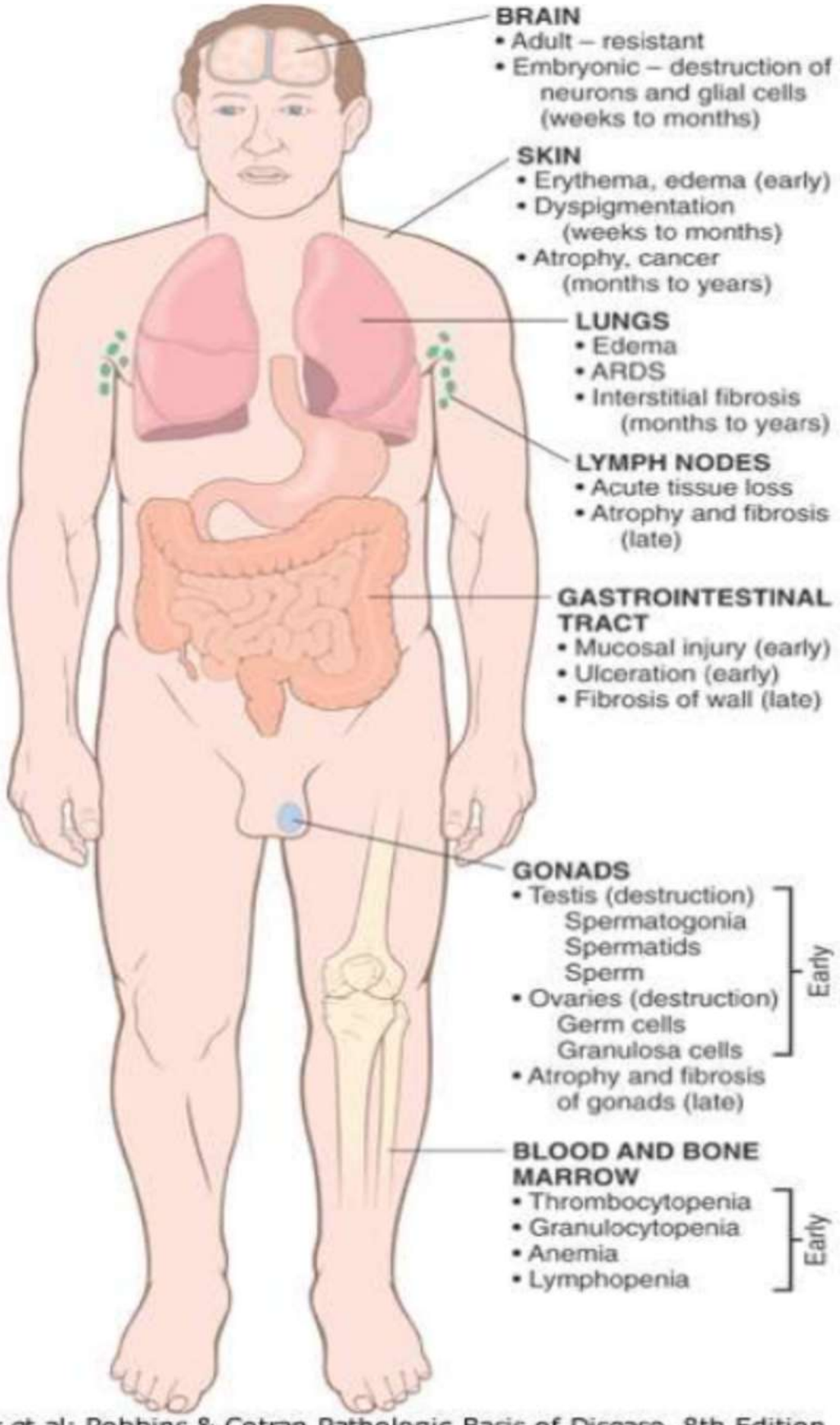
- Replacement of normal tissue by fibrosis
  - → scarring and loss of function
- Usually occur in ionizing radiation for cancer therapy
- Reminder: radiation causes cell injury → acute inflammation → chronic inflammation → repair by fibrosis

Carcinogenesis

- Hiroshima and Nagasaki survivors showed increased incidence of all types of leukemia
- Chernobyl survivors: increased *papillary thyroid cancer* (especially in children)
- Radiation may Induce genetic instability
  - Mutations persist in surviving cells
  - Accumulation of mutations from irreparable DNA lesions leads to **secondary cancers**, especially leukemias
- Radiation-induced cancers have LONG latency periods
  - Solid tumors appear after 10–20 years
  - Leukemias appear earlier (2–10 years)

- **Radiation injury affects tissues based on proliferative activity and vascular sensitivity.**
  Early effects: cell death in labile tissues (GI, bone marrow, skin).
  Late effects: fibrosis, atrophy, and increased cancer risk due to persistent DNA damage.

ARDS, Acute respiratory distress syndrome.



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Effects on growth and development

- Four susceptible phases

1. Preimplantation (0–2 weeks):
   - "All-or-none effect" (either lethal or no malformation)
2. Organogenesis (2–9 weeks)::
   - Highest risk for **major congenital malformations**
3. Fetal period (9 weeks to birth):
   - – CNS is the most radiosensitive → **mental retardation, decreased IQ, microcephaly**
4. Postnatal period:
   - Growth retardation; CNS, eyes, gonads highly sensitive

Ultraviolet radiation!!!

- UVA, UVB, UVC (UVB is the most carcinogenic component of sunlight.)
- Stratospheric ozone: important protective agent against UV
  - Absorbs all UVC, partially absorbs UVB
- Repeated exposure → premature aging
- UV light causes degeneration of collagen and elastin (contrast with ionizing radiation)
- UV induce DNA damage → malignant & premalignant skin lesions
- **Note:** UV is *non-ionizing*; DNA damage is *photochemical*

- Premalignant skin tumor
  - Actinic keratosis
- Malignant skin tumor
  - Squamous cell Ca
  - Malignant melanoma
  - Basal cell carcinoma